

1. When a key is pressed on a modern keyboard, what specific signal sequence does the internal microcontroller generate to send to the OS?

- (A) ASCII code
- (B) Make scan code
- (C) Break scan code
- (D) Unicode point

2. How does an optical mouse determine its displacement and direction of movement across a tracking surface?

- (A) By measuring change in magnetic capacitance
- (B) By capturing high-speed surface photos & comparing sequential frames
- (C) By calculating mechanical gear resistance
- (D) By measuring infrared time-of-flight distances

3. What standard font style is primarily used for printing MICR characters on banking checks to prevent fraud and ensure high accuracy?

- (A) Times New Roman
- (B) E-13B / CMC-7
- (C) Arial Black
- (D) OCR-A Bold

4. Which type of optical sensor component is typically employed inside high-resolution flatbed scanners to capture light reflections?

- (A) CCD (Charge-Coupled Device)
- (B) Thermistor array
- (C) Piezoelectric transducer
- (D) Hall effect sensor

5. What is the fundamental difference between an active stylus and a passive stylus used with touch screens and digitizers?

- (A) Active styluses contain internal power/logic for pressure levels
- (B) Passive styluses require Bluetooth pairing
- (C) Active styluses function only on optical infrared screens
- (D) Passive styluses send digitizer firmware updates

6. In keyboard design, "N-Key Rollover" (NKRO) refers to the capability of the keyboard to:

- (A) Automatically repeat keys at user-defined rates
- (B) Correctly register every key press regardless of how many are pressed simultaneously
- (C) Switch between QWERTY and DVORAK key mappings dynamically
- (D) Buffer up to N keystrokes during system latency

7. What issue occurs in cheaper keyboard matrices when pressing certain key combinations registers an unintended false key press?

- (A) Key Bouncing
- (B) Ghosting
- (C) Chattering
- (D) Latency Lag

8. In mouse specifications, what does the unit DPI (or CPI) directly measure?

- (A) Screen refresh rate in Hertz
- (B) Cursor steps reported to the OS per inch of physical movement
- (C) Data polling frequency per second
- (D) Sensor power consumption in milliwatts

9. Why is iron oxide or ferromagnetic material added to ink used in MICR check processing?

- (A) To make characters readable even if covered by stamp marks or signatures
- (B) To allow laser printers to print checks faster
- (C) To protect the paper from optical degradation
- (D) To enable UV light scanning for watermarks

10. Which touchscreen technology relies on measuring changes in electrical current when a conductive human finger touches the screen surface?

- (A) Resistive Touchscreen
- (B) Capacitive Touchscreen
- (C) Surface Acoustic Wave (SAW)
- (D) Infrared Grid Touchscreen

11. What is the main operational advantage of a Resistive Touchscreen over a Capacitive Touchscreen?

- (A) Better multi-touch response
- (B) Can be operated with any object (gloves, plastic stylus)
- (C) Higher optical clarity and brightness
- (D) Greater scratch resistance

12. What is the function of an Analog-to-Digital Converter (ADC) in voice input processing using a microphone?

- (A) Amplifies sound pressure waves
- (B) Converts continuous analog electrical signals into discrete binary speech samples
- (C) Filters background noise acoustically
- (D) Synthesizes text into human voice

13. What technology allows flatbed CIS (Contact Image Sensor) scanners to be more compact and energy-efficient than traditional CCD scanners?

- (A) Use of LEDs and a single-lens optical sensor array placed near paper
- (B) Elimination of glass plates in favor of thermal rollers
- (C) Use of high-power fluorescent tubes
- (D) Processing images directly without digital sampling

14. Which key switch mechanism provides tactile bumps and distinct clicky feedback through a mechanical spring and stem?

- (A) Rubber dome switch
- (B) Mechanical key switch
- (C) Membrane capacitive switch
- (D) Conductive strip switch

15. What term describes the rapid electrical oscillations produced when mechanical metal switch contacts touch, requiring circuit filtering?

- (A) Contact Bouncing
- (B) Crosstalk
- (C) Signal Jitter
- (D) Electromagnetic Interference

16. In standard 2D Barcodes like QR Codes, what feature allows information to be decoded accurately even if up to 30% of the code is damaged?

- (A) High-resolution scanning
- (B) Reed-Solomon Error Correction
- (C) Checksum bit verification only
- (D) Reflective ink encoding

17. In graphics digitizers, what metric measures the smallest physical distance change detectable by the stylus digitizing surface?

- (A) LPI (Lines Per Inch)
- (B) Bit Depth
- (C) Polling Hz
- (D) Frame Rate

18. Which input method uses infrared light beams forming an invisible grid over the display screen to detect touch coordinates without glass overlays?

- (A) Optical / Infrared Touch Matrix
- (B) Surface Acoustic Wave
- (C) Projected Capacitive
- (D) Electromagnetic Resonance

19. What happens when a user releases a key on a standard keyboard?

- (A) No signal is sent
(B) A Break scan code is transmitted
(C) The system resets the keyboard buffer
(D) An ASCII null character is sent

20. Which mouse metric specifies how many times per second the mouse reports its position data to the computer system?

- (A) DPI
(B) Polling Rate (Hz)
(C) Sensor Resolution
(D) IPS (Inches Per Second)

21. In document scanning, "Color Bit Depth" of 24-bit scanning indicates that:

- (A) The scanner uses 24 light sensors per inch
(B) Each pixel uses 8 bits for each primary RGB color channel (16.7 million colors)
(C) Images are compressed to 24KB per page
(D) The scanner operates at 24 frames per second

22. Electromagnetic Resonance (EMR) digitizer tablets (e.g., professional graphics tablets) operate by:

- (A) Emitting magnetic fields from the tablet loop array to power and track the battery-free pen
(B) Tracking optical reflection of lasers on glass surfaces
(C) Detecting static electricity accumulated on plastic nibs
(D) Measuring sound frequencies generated by pen tip friction

23. How does Optical Character Recognition (OCR) software transform image pixels into editable text characters?

- (A) By reading magnetic ink charges
(B) Through pattern matching and feature extraction algorithms
(C) By measuring darkness percentages on OMR grids
(D) By decoding barcode start-stop character sequences

24. Which peripheral interface standard introduced plug-and-play input device connectivity and replaced legacy PS/2 keyboard/mouse ports?

- (A) RS-232 Serial
(B) USB (Universal Serial Bus)
(C) IEEE 1284 Parallel
(D) VGA Interface

25. In modern high-performance gaming mice, what parameter represents the maximum physical speed at which the sensor can move without losing tracking?

- (A) IPS (Inches Per Second)
(B) DPI
(C) LPI
(D) PPM

ANSWER KEY

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(B)	6	(B)	11	(B)	16	(B)	21	(B)
2	(B)	7	(B)	12	(B)	17	(A)	22	(A)
3	(B)	8	(B)	13	(A)	18	(A)	23	(B)
4	(A)	9	(A)	14	(B)	19	(B)	24	(B)
5	(A)	10	(B)	15	(A)	20	(B)	25	(A)